

Error-Gated Endpoint Evaluation of Michell Resistance for Low-Froude B-Spline Hulls

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Abstract

Reliable low-Froude wave-resistance prediction requires both resolving an increasingly oscillatory Michell integral and estimating its algebraic tail. Michelsen’s 1960 dissertation and 1972 sequel developed analytic reductions of Michell resistance for polynomial and orthogonal-basis hull descriptions. We revive that lineage as a validated implementation for tensor-product B-spline hulls of degree at most 16 in each direction. Repeated integration by parts expresses the exact inner transform as endpoint waves. At low Froude number, an absolute bound permits the omission of terms below the waterline. Pair expansion then reduces resistance to kernels $K_s(\omega) = \int_1^\infty e^{i\omega\lambda} \lambda^{-s} (\lambda^2 - 1)^{-1/2} d\lambda$, analytically continued Bickley functions. An exact contour rotation converts each sufficiently separated nonzero-frequency pair to a Gaussian-decaying integral whose quadrature cost does not grow with frequency. The reported error includes coefficient construction, endpoint accumulation, omitted submerged pairs, and coarse/fine contour quadrature; the dispatcher falls back to a real-axis marcher when the estimate exceeds the requested tolerance. For the standard Wigley hull at $Fn = 0.02$, the method uses 1,152 kernel nodes instead of 511,504 inner-transform evaluations, reduces median runtime from 2.154×10^1 ms to 3.000×10^{-2} ms (about 700-fold), and differs from an independent real-axis reference by 1.83×10^{-13} relatively. The implementation is an engineering improvement for upright symmetric monohulls with coarse, well-separated active endpoints. Its contour-discretization estimate is not a formal interval certificate.

KEY WORDS: Wave resistance; Michell integral; low Froude number; oscillatory quadrature; B-splines; Bickley function.

Introduction

Michell’s 1898 thin-ship theory expresses wave resistance as a one-dimensional integral of the squared Fourier–Laplace transform of the longitudinal hull slope [Michell, 1898, Tuck, 1989]. Compared with nonlinear free-surface methods, Michell’s formula costs little and retains the bow–stern interference that couples resistance to speed and hull shape. Designers therefore use it for preliminary design, multihull studies, and mathematical shape optimization [Lazauskas, 2009, Dambrine et al., 2016]. At low speed, however, a fast result is useful only if its error estimate remains trustworthy.

An analytic reduction lineage addresses the same computational goal. Michelsen [1960], building on Birkhoff and Kotik’s transformation, splits the calculation into a hull function and a speed-dependent Michell function, derives a convergent special-function series for polynomial hull functions, and proposes tabulation for systematic design. Michelsen next treated polynomial centerline singularity distributions [Michelsen, 1963] and high- and low-speed asymptotic approximations [Michelsen, 1966]. Michelsen [1972] later uses Gegenbauer source distributions and orthogonality to reduce the integral to a finite double sum. The verified abstract of de Sendagorta and Grases [1988] likewise describes rapidly convergent series, products of shape functions, and shape-independent velocity functions for tabulation and computer-aided design. We treat this history as the method’s foundation and contribute an implementation-grade B-spline realization with measured error accounting and automatic fallback.

The apparent simplicity conceals a difficult numerical limit.

Let $\nu = g/U^2$, where U is speed and g gravitational acceleration. As U and therefore $Fn = U/\sqrt{gL}$ decrease, the phase $\nu\lambda x$ oscillates increasingly rapidly over the unbounded outer coordinate λ . Tuck used Filon-like longitudinal integration, while Lazauskas used exact piecewise-quadratic formulas and a fixed angular rule; Lazauskas reports that very low Froude number remains exceptional [Tuck, 1989, Lazauskas, 2009]. A real-axis adaptive marcher must spend more effort as frequency rises and must decide where to truncate the oscillatory algebraic tail. At the pre-fix revision 64925d4, one quiet-window criterion reported 5.123×10^{-9} relative error at $Fn = 0.05$, but the measured error was 5.781×10^{-7} , about 113 times larger. The frozen kernel advances quiet windows by oscillatory phase alone and adds the refinement and tail estimates. The four low-Froude calculations reported below all stop at the refinement cap: they are measured comparators, not claims that the requested 10^{-8} tolerance was achieved.

Earlier work also supplies the endpoint and contour ingredients. Wehausen [1973] surveys classical thin-ship theory and its limits. Keller and Ahluwalia show that bow and stern waterline data control the small-Froude wave field and resistance [Keller and Ahluwalia, 1976]. Gotman gives finite endpoint-derivative sums and products that separate bow, stern, and bow–stern interaction terms [Gotman, 2002, pp. 83–85]. Huybrechs and Vandewalle develop numerical steepest descent [Huybrechs and Vandewalle, 2006], and Motygin applies it to the Kelvin Green-function integral in linear ship-wave theory [Motygin, 2017]. We combine these ideas in four steps:

1. decompose every polynomial B-spline span exactly into

finitely many endpoint terms, including terms at interior and full-multiplicity chine knots;

2. bound every resistance pair that contains an omitted submerged endpoint;
3. identify the retained pair kernels as analytically continued Bickley functions and evaluate them on a frequency-scaled Gaussian contour; and
4. accept the reduced result only when its error estimate meets the requested tolerance; otherwise use the general real-axis method.

These are implementation deltas within established analytic and numerical traditions, not claims of method priority. The accessible publisher and index records establish the statements above for Michelsen [1972] and de Sendagorta and Grases [1988]; their full texts remain due-diligence items.

Michell resistance and spline geometry

Let $y = f(x, z) \geq 0$ denote the half-breadth of an upright symmetric hull, with $z \geq 0$ measured downward from the undisturbed waterline. In the normalization used by the implementation,

$$R_w = \frac{4\rho g^2}{\pi U^2} \int_1^\infty \frac{\lambda^2}{\sqrt{\lambda^2 - 1}} |F(\lambda)|^2 d\lambda, \quad (1)$$

$$F(\lambda) = \iint_\Omega f_x(x, z) e^{-\nu \lambda^2 z} e^{i\nu \lambda x} dx dz, \quad \nu = \frac{g}{U^2}. \quad (2)$$

All calculations below use $g = 9.807 \text{ m s}^{-2}$, freshwater density $\rho = 9.991 \times 10^2 \text{ kg m}^{-3}$, and the length Froude number $Fn = U/\sqrt{gL}$. We report

$$C_w = \frac{R_w}{\frac{1}{2}\rho U^2 S_w}, \quad S_w = 2 \iint_\Omega \sqrt{1 + f_x^2 + f_z^2} dx dz, \quad (3)$$

where S_w is the two-sided wetted area of the B-spline hull. Lengths, speeds, and forces are in metres, metres per second, and newtons; λ , Fn , and C_w are dimensionless. The hull is a clamped, non-rational tensor-product B-spline surface. On a knot rectangle $S = [x_0, x_1] \times [z_0, z_1]$, its longitudinal derivative is an exact bivariate polynomial,

$$f_x|_S = \sum_{a=0}^{p-1} \sum_{b=0}^q c_{ab}^S (x - x_0)^a (z - z_0)^b. \quad (4)$$

The method extracts these coefficients once from exact spline derivatives at span corners; it does not sample stations or waterlines. The implementation discards the zero-length intervals created by repeated knots. An interior knot whose multiplicity equals the degree represents a continuous chine exactly.

Exact endpoint decomposition

The finite polynomial degree makes repeated integration by parts terminate. For a polynomial p of degree m ,

$$\int_{x_a}^{x_b} p(x) e^{ikx} dx = \sum_{r=0}^m \frac{(-1)^r}{(ik)^{r+1}} \left[p^{(r)}(x) e^{ikx} \right]_{x_a}^{x_b}, \quad (5)$$

$$\int_{z_a}^{z_b} p(z) e^{-\kappa z} dz = - \sum_{u=0}^m \frac{1}{\kappa^{u+1}} \left[p^{(u)}(z) e^{-\kappa z} \right]_{z_a}^{z_b}. \quad (6)$$

Apply equations (5) and (6) to every monomial in equation (4), with $k = \nu\lambda$ and $\kappa = \nu\lambda^2$. The result is an exact endpoint representation.

Proposition 1 (Finite endpoint representation). *For a polynomial tensor-product B-spline hull with longitudinal degree at least one, the transform in equation (2) can be written*

$$F(\lambda) = \sum_{j=1}^M c_j \lambda^{-n_j} e^{i\nu \lambda x_j} e^{-\nu \lambda^2 z_j}, \quad n_j \geq 3, \quad (7)$$

where (x_j, z_j) are endpoints of nonzero knot spans and the complex coefficients c_j are finite combinations of span-polynomial endpoint derivatives. Equal endpoint/power triples may be combined exactly.

Proof. Equations (5) and (6) replace a monomial of derivative orders (r, u) by endpoint factors proportional to

$$(\nu\lambda)^{-(r+1)} (\nu\lambda^2)^{-(u+1)}.$$

The resulting power of λ is $n = r + 2u + 3$. Both sums terminate at the polynomial degrees, and summing the finite contributions from all nonzero knot rectangles gives equation (7). $\square$

Tests compare equation (7) directly with independent exact-moment calculations for both the Wigley hull and a multi-span hull with a full-multiplicity chine. Across $1 \leq \lambda \leq 20$, every discrepancy falls below the scaled 5×10^{-10} threshold.

Waterline reduction and omission bound

Partition equation (7) into waterline terms W , for which $z_j = 0$, and submerged terms S , for which $z_j > 0$. At low Froude number, every submerged term contains at least $e^{-\nu z_j}$ because $\lambda \geq 1$. Retaining only W gives

$$\tilde{F}(\lambda) = \sum_{j \in W} c_j \lambda^{-n_j} e^{i\nu \lambda x_j}. \quad (8)$$

Expanding $|\tilde{F}|^2$ before outer integration removes the nonanalytic complex conjugation from the integrand:

$$\tilde{R}_w = C \sum_{i, j \in W} \Re \{ c_i \bar{c}_j K_{s_{ij}}(\omega_{ij}) \}, \quad C = \frac{4\rho g^2}{\pi U^2}, \quad (9)$$

where $s_{ij} = n_i + n_j - 2 \geq 4$, $\omega_{ij} = \nu(x_i - x_j)$, and

$$K_s(\omega) = \int_1^\infty \frac{e^{i\omega\lambda}}{\lambda^s \sqrt{\lambda^2 - 1}} d\lambda. \quad (10)$$

Proposition 2 (Submerged-pair error bound). *Let $\mathcal{P}_S = \{(i, j) : i \in S \text{ or } j \in S\}$. The resistance discarded by replacing F with $\tilde{F}$ satisfies*

$$|R_w - \tilde{R}_w| \leq C \sum_{(i,j) \in \mathcal{P}_S} |c_i c_j| e^{-\nu(z_i + z_j)} K_{s_{ij}}(0). \quad (11)$$

Proof. Expand $|F|^2 - |\tilde{F}|^2$ into precisely the ordered pairs in $\mathcal{P}_S$. Apply the triangle inequality, use $|e^{i\omega\lambda}| = 1$, and bound $e^{-\nu\lambda^2(z_i + z_j)} \leq e^{-\nu(z_i + z_j)}$ for $\lambda \geq 1$. The remaining positive integral is $K_{s_{ij}}(0)$. $\square$

The zero-frequency kernel is analytic:

$$K_s(0) = \int_0^\infty \cosh^{-s} t \, dt = \frac{\sqrt{\pi} \Gamma(s/2)}{2\Gamma((s+1)/2)}, \quad (12)$$

A stable even/odd recurrence evaluates this kernel.

Bickley kernels and contour quadrature

With $\lambda = \cosh t$, [equation \(10\)](#) becomes

$$K_s(\omega) = \int_0^\infty e^{i\omega \cosh t} \cosh^{-s} t \, dt = \text{Ki}_s(-i\omega), \quad (13)$$

where the final equality denotes analytic continuation of the Bickley function [[Bickley and Naylor, 1935](#), [NIST Digital Library of Mathematical Functions, 2026](#)].¹ This identity supplies recurrences and other possible implementations. The present solver instead evaluates the contour directly.

The substitution $\lambda = 1 + t^2$ removes the square-root endpoint in [equation \(10\)](#):

$$K_s(\omega) = 2e^{i\omega} \int_0^\infty \frac{e^{i\omega t^2}}{(1+t^2)^s \sqrt{2+t^2}} dt. \quad (14)$$

For $\omega > 0$, analyticity in the intervening sector permits the exact rotation $t = e^{i\pi/4} r / \sqrt{\omega}$, giving

$$K_s(\omega) = \frac{2e^{i(\omega+\pi/4)}}{\sqrt{\omega}} \int_0^\infty \frac{e^{-r^2} dr}{(1+ir^2/\omega)^s \sqrt{2+ir^2/\omega}}. \quad (15)$$

Conjugation gives the negative-frequency kernels. The rotation replaces oscillation with Gaussian decay and leaves slowly varying frequency dependence in the amplitude. The same numerical-asymptotic principle yields frequency-independent work in high-frequency scattering [[Gibbs et al., 2020](#)]; here, the quadratic phase gives the contour in closed form.

For completeness, take the principal square root in [equation \(14\)](#). The integrand's poles $t = \pm i$ and branch points $t = \pm i\sqrt{2}$ lie outside the closed sector $0 \leq \arg t \leq \pi/4$. It is therefore analytic throughout the deformation. On a closing arc $t = Re^{i\phi}$, $|e^{i\omega t^2}| = e^{-\omega R^2 \sin(2\phi)} \leq 1$, while the algebraic factor including dt is $O(R^{-2s})$. The arc contribution vanishes as

¹DLMF calls Ki_s the “Bickley function” and cites the 1935 paper by W. G. Bickley and J. Naylor. The compound spelling “Bickley–Naylor” occurs in later literature, including [Ruffa and Toni \[2026\]](#); it is not DLMF terminology.

$R \rightarrow \infty$; thus the rotation is valid for every kernel order used here, $s \geq 4$.

The implementation normally applies 24- and 48-point Gauss–Legendre rules on $0 \leq r \leq 8$. When $s/|\omega| \geq 2$, the continued algebraic factor is narrow near the origin, so the code uses 48- and 96-point rules. The neglected Gaussian tail obeys

$$|K_s - K_s^{(8)}| \leq \frac{e^{-64}}{8\sqrt{2}\omega}, \quad \omega > 0, \quad (16)$$

using $|1 + ir^2/\omega| \geq 1$, $|\sqrt{2 + ir^2/\omega}| \geq \sqrt{2}$, and the standard Gaussian-tail bound. At the minimum supported nonzero frequency $\omega = 25$, this is below 4×10^{-30} . The selected coarse/fine difference estimates discretization error. This difference remains empirical. Thus the submerged-term bound and analytic contour-tail bound are rigorous, but the total floating-point error is not certified. A committed fixed-rule scan over $4 \leq s \leq 400$ and $|\omega| \in \{25, 100, 400\}$, using a 192-node same-contour reference and a 10^{-10} comparison floor, found a maximum relative 48-node discrepancy of 1.84×10^{-11} through $s = 50$, but 3.16×10^{-1} above that range. The 24/48 difference covered every meaningful 48/192 discrepancy by at least 2.22 in this scan. This measured degree dependence motivates the adaptive 48/96 rule within the validated $s \leq 98$ envelope; it is evidence, not a quadrature theorem.

For fixed geometry, a nonzero endpoint pair costs either 72 or 144 scalar nodes; increasing frequency never increases that cost. Self pairs use [equation \(12\)](#) without quadrature. The biquadratic Wigley representation has one longitudinal span and one vertical span. Its 16 endpoint terms include eight at the waterline. Their full ordered-pair expansion contains 32 nonzero-frequency entries; conjugacy and upper-triangular accumulation reduce these to 16 kernel evaluations. Each uses the 24/48 rule, for $16 \times 72 = 1,152$ scalar nodes in all; equal-position pairs use [equation \(12\)](#). The committed harness checks each count.

Hybrid algorithm and scope

The reduced method supplements the general solver. The following gates choose between them:

1. validate the physical conditions and exact spline geometry;
2. form and combine the exact endpoint terms in [equation \(7\)](#);
3. reject the specialization unless the hull is upright, symmetric, has its spline domain beginning exactly at $z = 0$, and has longitudinal degree at least one;
4. reject it if any distinct retained endpoint pair has $0 < |\omega_{ij}| < 25$;
5. evaluate [equation \(9\)](#), the bound [equation \(11\)](#), and the weighted coarse/fine contour difference; and
6. use the reduced result if their combined relative estimate meets the requested tolerance; otherwise run the general real-axis marcher.

The acceptance calculation is published here to make that last gate reproducible. If endpoint coefficient c_i combines a_i raw complex contributions q_{ik} by compensated accumulation, the implementation charges

$$\delta_i = \gamma_{2a_i+2} \sum_k |q_{ik}|, \quad \gamma_n = \frac{nu}{1-nu}, \quad (17)$$

where u is binary64 machine epsilon. For an upper-triangular retained pair (i, j) , let $m_{ij} = 1$ on the diagonal and 2 otherwise, and let K_{ij}^c and K_{ij}^f be the coarse and fine contour values. The three absolute resistance-level components are

$$B_o = C \sum_{(i,j) \in \mathcal{P}_S} |c_i c_j| e^{-\nu(z_i+z_j)} K_{sij}(0), \quad (18)$$

$$B_q = C \sum_{i \leq j} m_{ij} |c_i c_j| |K_{ij}^f - K_{ij}^c|, \quad (19)$$

$$B_r = C \sum_{i,j} (\delta_i |c_j| + |c_i| \delta_j + \delta_i \delta_j) e^{-\nu(z_i+z_j)} K_{sij}(0). \quad (20)$$

Here B_o is the omission bound in [equation \(11\)](#), B_q is the weighted contour-difference estimate, and B_r accounts for cancellation while constructing the endpoint map. A separate coefficient-construction floor is

$$\epsilon_c = \max(2 \times 10^{-8}, 4r_{\text{span}}), \quad (21)$$

where r_{span} is the independently sampled relative residual when the local power basis reconstructs f_x . Set $B = B_o + B_q + B_r$. If $B \geq |\tilde{R}_w|$, or any quantity is nonfinite, the reduced route refuses dispatch. Otherwise the printed and gated estimate is

$$\hat{\epsilon} = \epsilon_c + \frac{B}{|\tilde{R}_w| - B}. \quad (22)$$

The denominator follows from the reverse triangle inequality; using $B/|\tilde{R}_w|$ would understate an error relative to the unknown exact resistance. B_o , the separate Gaussian-tail inequality, and B_r under the standard floating-point model are analytic bounds. B_q and the validated floor ϵ_c are empirical estimates. Consequently $\hat{\epsilon}$ is an error-gating diagnostic, not a formal interval certificate.

The frequency gate reserves the fixed contour for sufficiently rapid oscillation. For the Wigley bow–stern pair, $\omega = \nu L = gL/U^2 = 1/Fn^2$, which equals the implementation’s $\nu|x_i - x_j|$. Thus $\omega \geq 25$ corresponds to $Fn \leq 0.2$. The same condition makes the geometry restriction explicit. For m equal longitudinal spans whose adjacent endpoints remain active, $\omega_{\text{adj}} = 1/(mFn^2)$, so the separation gate requires

$$m \leq \frac{1}{25Fn^2}. \quad (23)$$

At $Fn = 0.05$, for example, this permits at most 16 such spans. A retained multi-span endpoint map with 360 distinct nonzero-frequency pairs would use $360 \times 72 = 25,920$ ordinary-rule nodes, compared with 1,152 for the 16-pair Wigley map; this is node-count arithmetic, not a runtime benchmark. One closer endpoint invalidates the specialized path for the whole hull. The error gate

then bases the final choice on the actual hull, speed, and resistance cancellation rather than on nominal Froude number alone. At the default relative tolerance 10^{-5} , it rejects the Wigley reduction at $Fn = 0.08$ and accepts it at $Fn = 0.05$.

The implementation uses safe Rust and no external numerical library. Tests check the reduced solver at three independent levels: exact inner decomposition, isolated Bickley kernels, and total resistance. The broader validation harness checks Froude similarity, zero camber for symmetric sides, classical catamaran $4 \cos^2$ interference, tolerance self-convergence, and reported against observed error.

Numerical experiments

Independent reference

The standard Wigley half-breadth is

$$f(x, z) = \frac{B}{2} \left(1 - \frac{4x^2}{L^2}\right) \left(1 - \frac{z^2}{T^2}\right), \quad |x| \leq L/2, \quad 0 \leq z \leq T, \quad (24)$$

with $L = 1.000 \times 10^1$ m, $B = 1.000$ m, and $T = 6.250 \times 10^{-1}$ m. The primary reference shares neither the production moment code nor its outer integrator. It evaluates the analytic Wigley transform on the positive real axis after $\lambda = 1 + t^2$, which converts the outer weight to $2\lambda^2 |F(\lambda)|^2 / \sqrt{2+t^2}$. This explicitly removes the square-root singularity at $\lambda = 1$; [Tuck \[1989, p. 371\]](#) identifies that singularity and recommends its removal before quadrature.

Panels are first constructed in λ : $\Delta\lambda = \pi/(\nu L)$, so either endpoint phase $\nu\lambda L/2$ advances by at most $\pi/2$. Their endpoints are then mapped to t ; each panel uses a locally generated 16-point Gauss–Legendre rule, the cutoff is $\lambda_{\text{max}} = 4000$, and every nodal contribution is combined by Kahan accumulation. This requires 3,182,304 nodes at $Fn = 0.08$, rising to 50,916,864 at $Fn = 0.02$.

A second reference is independently transformed with $\lambda = \sec \theta$, Tuck’s stated regularization, for which the outer weight is $\sec^3 \theta |F(\sec \theta)|^2$. It maps the same phase-resolved λ endpoints into θ , constructs its own Gauss rule, and accumulates separately. [Table 1](#) separates endpoint map, panel-order, and cutoff convergence. The 8/16 trend and 16/24 agreement support the digits quoted below; floating-point agreement is still not an interval bound.

At $Fn = 0.02$, increasing λ_{max} from 4000 to 8000 changes the $1 + t^2$ and $\sec \theta$ references by 1.38×10^{-15} and 1.23×10^{-15} , respectively.

As a published-value anchor, [Doctors and Beck \[1987, Table 1\]](#) report the classical thin-ship value $10^3 C_w = 1.2486$ for $B/L = 0.1$, $T/L = 0.0625$, and $Fn = 0.35$. Our dimensional realization is the 1.000×10^1 m hull above, with $U = Fn\sqrt{gL}$, ρ and g as stated above, and C_w and S_w defined by [equation \(3\)](#). The committed harness gives $10^3 C_w = 1.247922$, a relative difference of 5.43×10^{-4} (0.0543%). We classify this agreement as a known result reproduced (class 1).

[Table 2](#) shows two regimes. At $Fn = 0.08$, submerged terms remain material, and the estimate rejects the reduction at default tolerance. At $Fn \leq 0.05$, reduced/reference differences range from 1.83×10^{-13} to 1.05×10^{-12} and remain within the reduced

Table 1: Independent-reference convergence. Entries are relative changes; “map” compares the 16-node $1 + t^2$ and $\sec \theta$ constructions. The final two columns compare panel orders within each construction.

Fn	Nodes/map	Map 16/16	t^2 8/16	t^2 16/24	$\sec \theta$ 8/16	$\sec \theta$ 16/24
0.08	3,182,304	1.284×10^{-15}	2.399×10^{-10}	1.426×10^{-16}	3.641×10^{-10}	0.000
0.05	8,146,704	3.280×10^{-16}	1.849×10^{-10}	9.839×10^{-16}	1.564×10^{-10}	1.640×10^{-16}
0.03	22,629,712	4.664×10^{-15}	1.364×10^{-10}	2.120×10^{-15}	1.420×10^{-10}	1.484×10^{-15}
0.02	50,916,864	1.534×10^{-15}	8.035×10^{-11}	6.136×10^{-15}	8.284×10^{-11}	5.523×10^{-15}

Table 2: Resistance validation against the primary analytic-Wigley reference. The marcher requested $\text{rel_tol} = 10^{-8}$ with six refinements, but every row reached RefinementCap (RC); no row is presented as tolerance converged. For $Fn \leq 0.03$, the physical omission correction is 6.2×10^{-32} or smaller, so the reduced/reference difference measures quadrature agreement rather than omission physics.

Fn	Reference R_w (N)	Reduced diff.	Reduced est.	Marcher diff.	Marcher est.	Marcher work	Status
0.08	1.946×10^{-1}	1.172×10^{-5}	3.817×10^{-5}	6.918×10^{-8}	2.896×10^{-7}	56,032	RC
0.05	1.058×10^{-2}	1.963×10^{-13}	2.000×10^{-8}	1.493×10^{-7}	5.974×10^{-7}	119,936	RC
0.03	5.114×10^{-4}	1.045×10^{-12}	2.000×10^{-8}	3.322×10^{-7}	1.320×10^{-6}	267,856	RC
0.02	4.417×10^{-5}	1.829×10^{-13}	2.000×10^{-8}	6.334×10^{-7}	2.517×10^{-6}	511,504	RC

solver’s empirical estimates. The general marcher’s error, by contrast, grows as Froude number falls. Its estimate covers the observed error in all four rows, but each calculation exhausted the configured refinement budget. The independent endpoint-aware references, not the capped marcher, provide the comparison standard in [table 2](#).

The isolated kernel tests compare [equation \(15\)](#) against dense real-axis integration for $s \in \{4, 7, 10, 50, 98, 128\}$ and $|\omega| \in \{25, 100, 400\}$, with scaled discrepancy below 10^{-9} . The public degree envelope reaches at most $s = 98$; order 128 provides margin. The full Rust and Python-interface suites pass.

Runtime

Timing used `cargo bench -p mitchell --bench wigley`, 30 samples per case, the optimized Cargo bench profile, no custom RUSTFLAGS, and the default requested relative tolerance 10^{-5} . Each case received an unmeasured warm call. Calls below 1.000×10^{-1} ms were timed in batches of 256 and divided by 256; checksums use separate untimed calls, so timer and checksum work do not overlap. The two method comparisons alternated execution order on every sample; other cases were sampled separately in printed order. The host was a 10-core Apple M5 MacBook Air with 24 GB RAM, arm64 Darwin 25.5.0, Rust/Cargo 1.96.0, and LLVM 22.1.2. Absolute timings are host-dependent; work counts and checksums are deterministic.

At $Fn = 0.02$, the frozen paired run gives a 718-fold median speedup; repeated protocols on this host have spanned roughly 608–865-fold, so the ratio is not portable. The comparison uses equal *requested* tolerance, not matched observed accuracy: the endpoint/reference difference is 1.83×10^{-13} , whereas the capped marcher’s is 6.33×10^{-7} . Cost changes little between $Fn = 0.05$ and 0.02, as [equation \(15\)](#) predicts. The sweep grid is the 21 speeds $Fn = 0.10, 0.12, \dots, 0.50$; its checksum is the sum of the 21 returned resistances from one untimed pass. The scalar low-speed checksums are likewise sums of one result per sample, outside the timed batches. During the sweep, the hybrid selects the

path that satisfies the error gate. The checksum differs from the pre-fix value because the corrected marcher retains a positive tail that the earlier rule truncated.

Analytic lineage and implementation scope

The present solver revives a substantial analytic lineage. The equation-level comparison in [table 5](#) separates the historical reductions from the implementation work reported here. For the 1972 and 1988 papers, the table states only what their verified publisher or index records establish; both full texts remain due-diligence items rather than dependencies of our classification.

The implementation differs from that lineage in five engineering details:

1. it operates span by span on tensor-product B-splines, including full-multiplicity chine knots;
2. it restricts both spline degrees to the independently validated range 1–16 and charges coefficient construction and endpoint cancellation;
3. it bounds the resistance contribution of every omitted submerged endpoint pair;
4. it evaluates retained K_s pairs on a frequency-scaled contour with a bounded, order-aware node count; and
5. it accepts the reduction only when the combined estimate meets the requested tolerance, otherwise returning to the general marcher.

These deltas are engineering improvements, not priority claims. A full method-by-method benchmark against the Michelsen and Sendagorta–Grases representations lies outside this paper and remains future work.

The endpoint structure is also established independently. [Keller and Ahluwalia \[1976\]](#) express leading small-Froude resistance and waves through bow and stern waterline slopes. [Gotman](#)

Table 3: Decomposition of the reduced estimate in [equation \(22\)](#). The first three columns use the guarded denominator $|\tilde{R}_w| - B$; their sum plus the construction/safety floor is the printed estimate.

F_n	Omission B_o/D	Contour B_q/D	Rounding B_r/D	Floor ϵ_c	Total
0.08	3.815×10^{-5}	1.355×10^{-12}	9.153×10^{-15}	2.000×10^{-8}	3.817×10^{-5}
0.05	3.656×10^{-12}	7.222×10^{-14}	9.702×10^{-15}	2.000×10^{-8}	2.000×10^{-8}
0.03	6.153×10^{-32}	3.462×10^{-12}	9.292×10^{-15}	2.000×10^{-8}	2.000×10^{-8}
0.02	5.508×10^{-70}	7.620×10^{-13}	9.428×10^{-15}	2.000×10^{-8}	2.000×10^{-8}

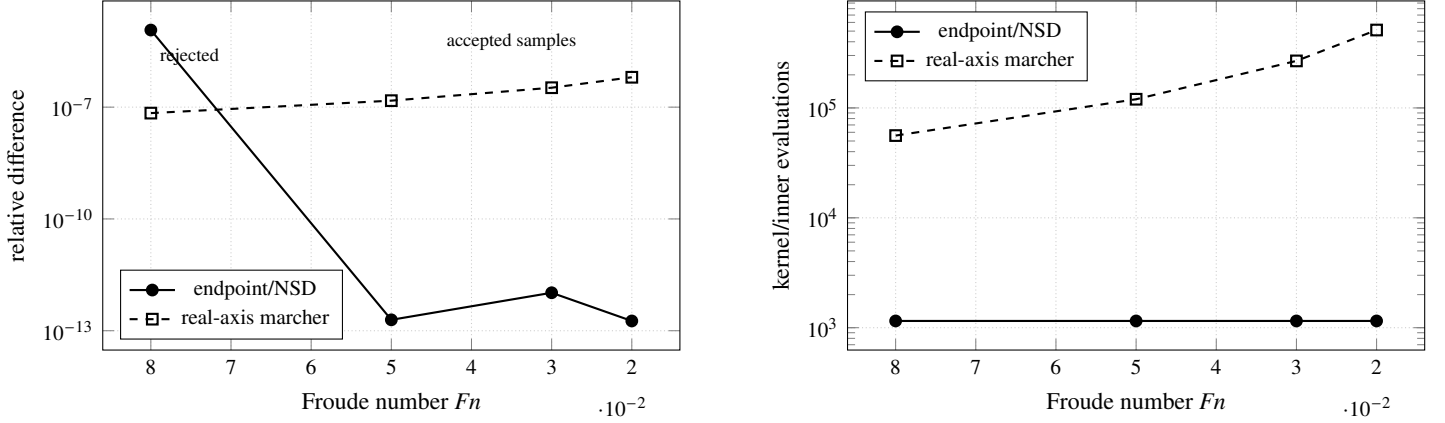


Figure 1: Observed accuracy (left) and work (right) as Froude number decreases. The two endpoint-aware references agree in the low 10^{-15} range; the plotted endpoint/NSD differences are therefore resolved quadrature comparisons. Every real-axis marcher point has status `RefinementCap`. At default tolerance, the hybrid rejects the reduced result at $F_n = 0.08$ and accepts it at $F_n = 0.05, 0.03, 0.02$.

[2002, pp. 83–85] derives finite endpoint-derivative sums, forms their products in resistance, and separates bow, stern, and bow–stern contributions; this is closer to the present pair structure than a generic integration-by-parts citation. Our finite B-spline decomposition puts that mechanism into implementation form; it does not introduce a new physical asymptotic principle.

Tuck and Lazauskas also establish exact or Filon-like integration of piecewise-polynomial inner data. In particular, [Lazauskas \[2009, sec. 6.5, pp. 6–10–6–12\]](#) gives exact piecewise-quadratic hull integrals, compares equally spaced Simpson and cosine-spaced trapezoidal angular rules, and reports that very low Froude number remains exceptional. The accessible thesis supports those numerical practices but does not, by itself, document the internals of every Michlet release. Our solver retains the same thin-ship physics but, for accepted B-spline hulls, keeps low-Froude outer work from growing with frequency. We classify that result as an engineering improvement and make no head-to-head performance claim against the Michlet executable. Exact endpoint reduction also survives the outer modulus square: pair expansion converts it into a small set of reusable special-function kernels. Ruffa and Toni give finite Bessel–Struve module representations for integer-order Bickley functions [[Ruffa and Toni, 2026](#)]; their stability on the imaginary axis remains to be tested as a possible kernel backend.

Contour deformation is well established for oscillatory quadrature [[Huybrechs and Vandewalle, 2006](#)]. [Motygin \[2017\]](#) applies steepest descent and Clenshaw–Curtis quadrature in ship-wave theory, but to the oscillatory part of the Kelvin Green function

rather than Michell’s resistance integral. Automated methods now handle general phases and coalescing saddles [[Gibbs et al., 2024](#)]. Such methods could treat multihull transverse phases; the present quadratic phase needs only the closed contour in [equation \(15\)](#).

Limitations and open extensions

The evidence covers a narrow model: upright symmetric monohulls in linear, deep-water thin-ship theory.

- **Geometry.** The reduced solver supports upright symmetric single hulls of degree 1–16 in each spline direction. Its active endpoints must satisfy $v|x_i - x_j| \geq 25$, so it targets coarse, well-separated endpoint maps. Asymmetric, heeled, multihull, and closely spaced cases fall back to the general method.
- **Contour degree.** The fixed 48-node rule was accurate through $s \approx 50$ and degraded beyond it in scans to $s = 400$; its 24/48 difference nevertheless remained conservative against the 192-node same-contour comparator throughout that scan. Production therefore switches to 48/96 when $s/|\omega| \geq 2$, and the claimed envelope ends at the independently tested public maximum $s = 98$.
- **Error certification.** The submerged-pair omission bound and Gaussian tail bound are analytic. The selected coarse/fine quadrature difference and floating-point round-off are not interval-certified.

Table 4: Final release benchmark on the development host.

Case	Median (ms)	IQR (ms)	Diagnostic
21 speeds, $F_n = 0.10 \dots 0.50$	13.059	12.952–13.126	checksum 2.553251156101 $\times 10^5$
Default API, $F_n = 0.05$	0.030	0.030–0.030	1,152 nodes; batch 256
Real-axis marcher, $F_n = 0.02$	21.541	21.311–21.666	511,504 evaluations; batch 1
Endpoint/NSD, $F_n = 0.02$	0.030	0.030–0.030	1,152 nodes; batch 256

- **Reference floor.** The independent $1 + t^2$ and $\sec \theta$ references use compensated summation through $\lambda = 4000$. At $F_n = 0.02$, their cutoff-doubling changes are 1.38×10^{-15} and 1.23×10^{-15} , respectively. Differences near the observed 10^{-13} scale are resolution measurements, not interval-certified digits.
- **Physical model.** Michell theory is linear, inviscid, slender, deep-water, and fixed-attitude. At very low speed, viscous effects may dominate the already small wave resistance. For free-surface flow past singular obstacles, low-Froude waves can be exponentially small, beyond all algebraic orders, and controlled by Stokes phenomena; that different problem is not resolved by better quadrature [Trinh and Chapman, 2015].

The highest-leverage numerical extension is multihull support. A transverse placement introduces phase $\nu y \lambda \sqrt{\lambda^2 - 1}$ and moving saddles that equation (15) does not contain. A uniform or automated contour must account for them. Retaining shallow submerged endpoints likewise produces mixed linear and quadratic complex phases. Further work should certify the contour error, test imaginary-argument Bickley recurrences, derive finite-depth kernels, and differentiate the endpoint representation for low-Froude hull optimization.

Conclusions

Finite integration by parts converts the B-spline inner transform into endpoint waves. Pair expansion then turns the outer modulus square into continued Bickley kernels, and contour rotation replaces real-axis oscillation with Gaussian decay. Geometry, rather than Froude number, sets the reduced quadrature work. The submerged-pair bound lets the hybrid reject the reduction whenever it exceeds the requested tolerance.

For the standard Wigley hull at $F_n = 0.02$, the reduced solver is more accurate than the general marcher and about 700 times faster in the frozen paired run. This result is an engineering improvement for the supported geometry, built from the analytic reduction lineage summarized in table 5. Claims beyond the current scope require broader geometry and certified total error control.

Reproducibility statement

The implementation, tests, benchmark harness, generated results, and number audit are available in the public [michell repository](#). On acceptance for submission, the immutable source, test, and

artifact package will be archived and the placeholder [10.5281/zenodo.REPLACE-ME](#) replaced with its minted DOI. The current numerical kernel will be identified by the immutable local tag `paper-a-jsr-v3`; that tag will not be moved after creation.

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Table 5: Equation-level comparison of polynomial and basis reductions of Michell resistance. “Record” denotes a verified publisher or bibliographic record where the full text was unavailable.

Source	Basis and reduction	Kernel or speed dependence	Error control and dispatch	Relation to this implementation
Birkhoff–Kotik via Wehausen [1973, eqs. 39–45]	Hull autocorrelation $M(u, v)$, with equivalent (P, Q) transforms and separable elementary ships	Geometry-independent kernel $K(vu, vv)$; a Y_0 reduction is also reported	Absolute convergence justifies the change of integration order	Establishes separation of hull data from a reusable kernel
Michelsen [1960, eqs. 3.3–3.4]	Polynomial hull function $H(\xi, \zeta)$ after the Birkhoff–Kotik transformation	Michell function $C(s, t)$ contains speed dependence	Convergence conditions accompany the transformation	Direct historical antecedent of shape/kernel separation
Michelsen [1960, eqs. 3.21–3.22, 3.38–3.40]	Monomial expansion of H ; each coefficient contributes through a closed series expression	Confluent-hypergeometric, Bessel, and Struve functions; speed-and-degree terms can be tabulated	Appendix proves series convergence; computer tabulation is proposed	Antecedent of analytic polynomial reduction for design use
Michelsen [1963], program record	Polynomial centerline singularity distributions	Direct evaluation of their wave-resistance contribution	The verified seminar program establishes topic and venue; detailed controls await source access	Connects the dissertation’s polynomial program to centerplane distributions
Michelsen [1966], bibliographic record	High- and low-speed asymptotic approximations to Michell’s integral	Speed-limit formulas rather than a general finite-speed kernel	The verified journal record establishes scope; detailed remainder control awaits source access	Extends the lineage explicitly into both speed limits
Michelsen [1972], verified record	Gegenbauer expansion of the centerplane Havelock-source distribution	Orthogonality reduces resistance to a finite double sum depending on source coefficients, F_n , and length–draft ratio	The accessible abstract states the reduction and identifies its governing inputs	Published JSR continuation of the basis-reduction program
de Sendagorta and Grases [1988], verified record	Products of integral shape functions	Rapidly convergent series with linear, shape-independent velocity functions suitable for tabulation	The abstract reports that few terms give suitable accuracy; detailed controls await source access	Antecedent of separated, table-driven computer-aided design
Present implementation	Degree- ≤ 16 tensor-product B-spline span polynomials reduced to active endpoint waves	Continued Bickley pair kernels on a frequency-scaled Gaussian contour	Coefficient, accumulation, omission, and quadrature estimates; explicit acceptance gate and real-axis fallback	Modern error-accounted realization for coarse, well-separated endpoints

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